

# Direct Collisional Method of Molecular Dynamics Simulation

## Senior Thesis Defense

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## 1. Introduction & Background

- The Computational Challenge
- Rarefaction & Knudsen Number
- Traditional Methodologies (MD vs. DSMC)

## 2. Theoretical Framework

- Proposed Direct Collisional Model
- Algorithmic Execution Flow
- Collision Mechanics in the CMF

## 3. Simulation & Results

- System Configuration Setup
- Ergodic Velocity Distributions
- Quantitative Verification ( $T = 343$  K)

## 4. Synthesis

- Application: Stopping Power
- Future Model Extensions
- Concluding Remarks



# The Computational Challenge

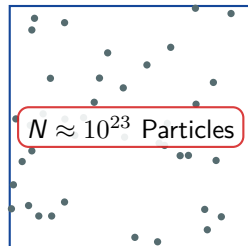
## The Scaling Problem

- Real-world gaseous systems contain  $\approx 10^{23}$  particles.
- Tracking every trajectory and force interaction is **computationally prohibitive**.

## Theoretical Foundations

- **Equilibrium:** Follows the Maxwell-Boltzmann Distribution (MBD).
- **Non-Equilibrium:** Governs critical scenarios like aerospace atmospheric re-entry.

### Macroscopic Regime



# When Do Standard Fluid Equations Fail?

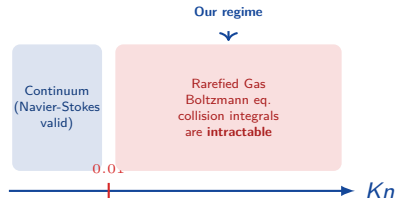
## The Core Motivation

Whenever the mean free path  $\lambda$  is **comparable to the system size**  $L$ , individual molecular collisions dominate — and continuum fluid equations **break down entirely**.

## The Knudsen Number

$$Kn = \frac{\lambda}{L}$$

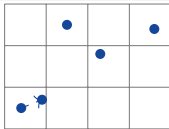
- $Kn > 0.01$ : The system is **rarefied** — particle-level kinetics must be resolved.
- Examples: high-altitude flight, vacuum systems, microfluidic devices.



# Existing Methods: Why Neither is Enough

## Direct Simulation Monte Carlo (DSMC)

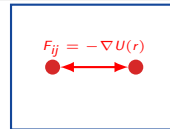
- Probabilistic collisions inside spatial grid cells.
- **Fast:**  $\mathcal{O}(N_p)$  scaling.
- **Gap:** No continuous particle tracking — indirect physical picture.



Grid Cells (Probabilistic)

## Classical Molecular Dynamics (MD)

- Deterministic Newton's equations for all pairs.
- **Accurate:** Full trajectory history preserved.
- **Gap:** Catastrophic  $\mathcal{O}(N_p^2)$  scaling — unaffordable at scale.



Continuous Force Field

**Goal:** Capture the physical accuracy of MD at the computational cost of DSMC.



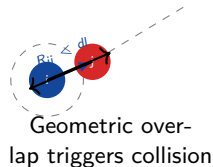
## Hybrid Design Principles

- Blends the **efficiency of DSMC** with the **direct particle representation of MD**.
- Employs a rigid, deterministic **Hard-Sphere Potential**:

$$\phi(r) = \begin{cases} \infty, & r \leq dl \\ 0, & r > dl \end{cases}$$

- Trajectories propagate freely:  $d\mathbf{X} = \mathbf{V} dt$ .
- **No continuous force fields** — collision detected geometrically.

## Collision Condition



# The Simulation Algorithm

1. **Initialize Phase Space:**  $N \times 3$  arrays for Position  $\mathbf{X}$  and Velocity  $\mathbf{V}$  (from MBD)

2. **Free Propagation:** Update positions by  $d\mathbf{X} = \mathbf{V} dt$

3. **Overlap Check:** Evaluate pairwise separation  $R_{ij}$

4. **Execute Collisions:** Apply CMF scattering if  $R_{ij} < dl$

5. **Boundary Conditions:** Enforce periodic boundaries + save state

Repeat until  $t_{end}$



## Kinetic Transformation

Transform to the Center of Mass Frame (CMF):

$$\mathbf{v}_1 = \mathbf{V}_c - \frac{\mu}{m_1} \mathbf{v}_r, \quad \mathbf{v}_2 = \mathbf{V}_c + \frac{\mu}{m_2} \mathbf{v}_r$$

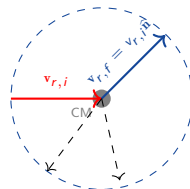
$\mu = \frac{m_1 m_2}{m_1 + m_2}$  is the reduced mass.

## Conservation Laws

$$KE_T = \frac{1}{2} M V_c^2 + \frac{1}{2} \mu v_r^2$$

- $\mathbf{V}_c$  is invariant (no external forces).
- Elastic scattering: conserve  $|\mathbf{v}_r|$ .
- Scattering direction  $\hat{\mathbf{n}}$ : uniform random on  $4\pi$ .

## Isotropic Scattering in CMF



Uniform  $4\pi$  angular probability





# Preventing Artificial Double-Collisions

## The Problem

A small  $dt$  can leave overlapping particles still within  $R_{ij} < dl$  in the next step, triggering **non-physical double-collisions**.

## The Fix: Spatial Displacement

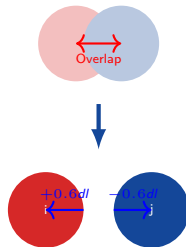
After each collision, manually separate the pair along  $\hat{\mathbf{R}}$ :

$$\mathbf{X}_i \leftarrow \mathbf{X}_i + 0.6 \, dl \, \hat{\mathbf{R}}$$

$$\mathbf{X}_j \leftarrow \mathbf{X}_j - 0.6 \, dl \, \hat{\mathbf{R}}$$

Guarantees geometric clearance before the next loop.

Non-physical:  $R_{ij} < dl$  persists

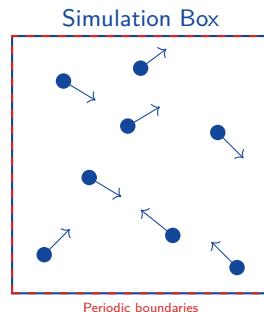


# Simulation Setup

## Physical Parameters

Conditions represent a low-density atmospheric gas:

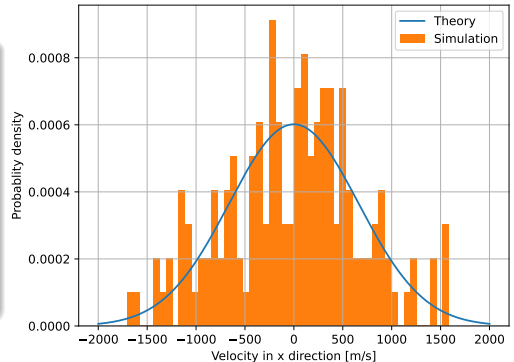
| Parameter          | Symbol | Value                                |
|--------------------|--------|--------------------------------------|
| Particle Count     | $N_p$  | 150                                  |
| Number Density     | $n$    | $2.34 \times 10^{24} \text{ m}^{-3}$ |
| Temperature        | $T$    | 300 K                                |
| Molecular Diameter | $d_l$  | $4.0 \times 10^{-10} \text{ m}$      |
| Box Volume         | $V$    | $6.4 \times 10^{-23} \text{ m}^3$    |
| Particle Mass      | $M$    | $10^{-26} \text{ kg}$                |



# Results: Velocity Distribution

## Ergodicity and Equilibrium

- A single particle tracked over  $1.7 \mu\text{s}$  ergodically samples the full velocity distribution.
- The simulated histogram for  $v_x$  aligns closely with the theoretical Maxwell-Boltzmann curve at  $T = 318 \text{ K}$ .
- Small fluctuations reflect finite-time statistical noise.



## Theoretical Verification Table

| Physical Metric                              | Computed Mean                      | Theoretical Mean                   | Ratio        |
|----------------------------------------------|------------------------------------|------------------------------------|--------------|
| Mean Particle Energy $\langle E \rangle$     | $1.927 \times 10^{-21}\text{ J}$   | $2.367 \times 10^{-21}\text{ J}$   | <b>0.814</b> |
| Specific Heat Capacity $\langle c_v \rangle$ | $5.388 \times 10^{-24}\text{ J/K}$ | $6.900 \times 10^{-24}\text{ J/K}$ | <b>0.781</b> |
| Mean Free Path $\lambda$                     | $1.216 \times 10^{-6}\text{ m}$    | $6.001 \times 10^{-6}\text{ m}$    | <b>0.203</b> |

- $\langle E \rangle$  and  $\langle c_v \rangle$ : within  $\sim 20\%$  of theory — excellent agreement.
- $\lambda$  discrepancy: hard-sphere packing effects vs. dilute ideal-gas assumption.
- All results within the same order of magnitude — model validated.



# Application: Radiation Stopping Power

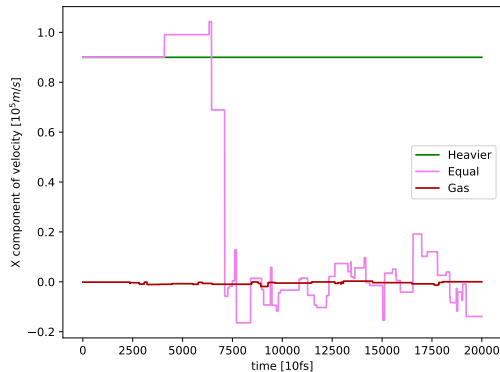
## Maximum Energy Transfer

An energetic particle  $K_{in}$  injected into the gas loses energy via:

$$T_{max} = \frac{4m_1m_2}{(m_1 + m_2)^2} K_{in}$$

| Case              | Transfer fraction     | Behavior      |
|-------------------|-----------------------|---------------|
| $m_1 \approx m_2$ | $\approx 1.0 K_{in}$  | Rapid stop    |
| $m_1 = 2m_2$      | $\approx 0.89 K_{in}$ | Moderate loss |
| $m_1 \gg m_2$     | $\approx 0$           | Straight path |

- Simulation reproduces both extremes and intermediate cases correctly.



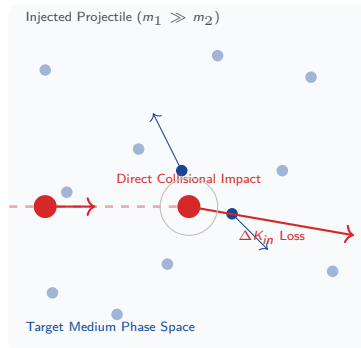
## Identified Limitations

- Hard-sphere model restricted to **dilute** (low-density) regimes.
- Finite simulation time causes minor statistical fluctuations in  $\langle E \rangle$  and  $\langle c_v \rangle$ .
- No long-range molecular forces included.

## Proposed Extensions

- **Longer runs** to fully converge statistical moments.
- **External fields:** Add gravity or EM force via deterministic velocity shifts.
- **Soft potentials:** Replace hard core with Lennard-Jones for dense-gas regimes.

## Stopping Power Kinematics



## What We Built and Validated

- A **Direct Collisional hybrid model** combining DSMC efficiency with MD physical accuracy.
- Results for  $\langle E \rangle$ ,  $\langle c_v \rangle$ , and  $\lambda$  are within the same order of magnitude as theory.
- Verified convergence to the **Maxwell-Boltzmann distribution** via ergodic tracking.
- Model flexibility demonstrated through radiation stopping power simulations.

The Direct Collisional Method offers a **practical, scalable** path for simulating rarefied gas dynamics where classical methods fail.





# Thank You for Your Attention!

Questions & Comments are Welcome.

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## Key Literature

- ① **Bird, G. A.** (1994). *Molecular Gas Dynamics and the Direct Simulation of Gas Flows*. Clarendon Press, Oxford. \*(Foundational reference for DSMC scaling mechanics rules)\*.
- ② **Rapaport, D. C.** (2004). *The Art of Molecular Dynamics Simulation*. Cambridge University Press. \*(Core framework utilized for classical deterministic tracking algorithms)\*.
- ③ **Alder, B. J., & Wainwright, T. E.** (1959). Phase Transition for a Hard Sphere System. *Journal of Chemical Physics*, 31(2), 459–466. \*(The basis for the rigid potential condition implemented in our code)\*.
- ④ **Cercignani, C.** (1988). *The Boltzmann Equation and its Applications*. Springer New York. \*(Theoretical basis for non-equilibrium kinetic regimes)\*.

